

Bus Companies

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities and m bus companies. Each bus company operates in specific cities and sells tickets for a specific price. Buying a ticket from a bus company allows you to travel between any two cities that the company operates in.

Determine the cost of the cheapest route from Syrjälä to every city.

Input

The first line has two integers n and m : the number of cities and bus companies. The cities are numbered $1, 2, \dots, n$, and city 1 is Syrjälä.

The next line has m integers $c_1, c_2, \dots, c_m$: the ticket costs for each bus company.

After that, there are m pairs of lines describing the cities for each bus company.

The first line of each pair has a single integer k : the number of cities the bus company operates in.

The second line of each pair has k distinct integers $a_1, a_2, \dots, a_k$: the cities the bus company operates in.

You can assume that it is possible to travel from Syrjälä to all other cities.

Output

Print n integers: the cheapest route costs from Syrjälä to cities $1, 2, \dots, n$.

Constraints

- $1 \leq n, m \leq 10^5$
- $1 \leq c \leq 10^9$
- $2 \leq k \leq n$
- $1 \leq a \leq n$
- the sum of all k is at most $2 \cdot 10^5$

Example

Input	Output
5 3 4 3 2 3 1 4 3 2 5 1 4 2 3 4 5	0 5 4 4 3